# Weihang (Charles) Ding

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## Research Interests

LLM and agent evaluation (benchmark design, LLM-as-a-judge reliability and bias); post-training and reliable delivery (SFT, RL, agents as forward-deployed engineers); efficient AI systems (cost-aware model routing, dynamic model escalation, quantized inference).

## Education

**University of California, Berkeley** Aug. 2025 – Aug. 2026
*Master of Analytics (Machine Learning), College of Engineering* *GPA 3.86/4.0*

**University of Birmingham** Sep. 2021 – Jun. 2025
*BSc in Applied Mathematics with Information Computing Science, First Class Honours* *GPA 4.0/4.0*

- Chair of the Staff–Student Committee.

## Publications

[1] **Weihang Ding** and Junfei Zhan. "Trains but Doesn't Learn: A Post-Training Delivery Benchmark for LLM Agents as Forward-Deployed Engineers." In *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing (EMNLP): Industry Track*, pages 868–879, Budapest, 2026. [paper]

[2] **Weihang Ding**, Junfei Zhan, Yueting Li, and Qirong Guo. "FDE-Bench: Evaluating LLM Agents for Deployment Environment Configuration." *NeurIPS 2026 Workshop on Agentic AI Benchmarks and Applications for Enterprise Tasks (AABA4ET)*, Sydney, 2026. [paper] [workshop]

[3] **Weihang Ding** and Junwei Duan. "BroadDiff: A Novel Multimodal Broad Diffusion Model for Early Diagnosis of Alzheimer's Disease." *Biomedical Signal Processing and Control*, 2026. [paper]

[4] **Weihang Ding**, Yuxin Luo, Yifeng Lin, Yuer Yang, and Siwei Lian. "VeriBypasser: An Automatic Image Verification Code Recognition System Based on CNN." *Computer Communications*, 217:246–258, 2024. [doi]

[5] Yueting Li, **Weihang Ding**, and Meijia Chen. "SGAnalog: An End-to-End Circuit Benchmark from Open-Source Silicon Tapeouts." *NeurIPS 2026 Workshop on AI for Chip Design*, 2026. **Oral**. [paper] [openreview]

[6] **Weihang Ding** et al. "AdaFleet: An Adaptive Orchestrator–Subagent Coevolution Framework for Efficient Quantitative Signal Discovery." Under review at *ICLR 2027*.

**Technical reports**

[7] **Weihang Ding**. "Dynamic Model Escalation for Coding Agents: A GMI Router Experiment Across Two Harnesses." GMI Cloud technical blog, Sep. 2026. [post]

## Research Experience

**Stanford Graduate School of Business, AIM-Lab** Oct. 2025 – May 2026
*Research Assistant* *Stanford, CA*

- Built a high-concurrency LLM benchmarking pipeline evaluating 7 models over 300K+ API calls in a 3-month longitudinal study under varied prompt constraints.
- Measured LLM-as-a-judge reliability: quantified position, verbosity, and self-preference bias in automated scoring and evaluated debiasing methods for each effect.
- Built trustingdata.ai, a full-stack research platform on Google Cloud supporting Stanford GSB and Wharton studies of human–LLM and human–agent interaction.

**University of Birmingham** Nov. 2022 – Apr. 2025
*Undergraduate Researcher (lead author on two journal papers)*

- **BroadDiff**: combined graph diffusion with the Broad Learning System to capture long-range node information while avoiding the overfitting of graph convolution; validated on multi-stage Alzheimer's disease classification. [Pub. 3]
- **GDRBN**: Graph Diffusion Regularized Broad Network with Hippo-algorithm hyperparameter search for visual recognition on ImageNet (Oct. 2024 – Apr. 2025).
- **VeriBypasser**: led a 5-person team to build a CNN-based CAPTCHA recognition system (85.2% accuracy) and a CAPTCHA generator with controllable distortion; assessed the risk of weak CAPTCHA schemes. [Pub. 4]

## Industry Experience

**GMI Cloud (NVIDIA Preferred Partner)** May 2026 – Present
*Machine Learning Engineer* *Mountain View, CA*

- Led the GMI Router, an LLM router predicting per-query quality and cost across models. Against GPT-5.5 alone, balanced mode cuts cost 79% at equal quality and quality mode cuts cost 31% at 2.5% higher quality; both score over 10% above the OpenRouter Auto Router.
- Designed the dynamic model escalation experiment on the Terminus 2 and Claude Code harnesses: start on a 26× cheaper model, escalate on live failure signals, and carry the transcript and working tree across the switch. Recovered 3 of 9 Terminal-Bench 2.1 tasks that had failed in all 10 prior runs.
- Showed that the harness, not only the task, decides which tasks need the stronger model (0.35 Jaccard overlap across harnesses), making harness identity a routing input.
- Built a 12-benchmark, 21-model evaluation suite with 90K+ graded responses using rule checkers, sandboxed code execution, and an LLM-judge panel.
- Deployed an NVFP4-quantized Nemotron-3-1B embedding model on 8× Blackwell 5090 GPUs with vLLM: 3× throughput at 1.5M input tokens/s with no observable quality loss.

**Tencent (the company behind WeChat)** Apr. 2025 – Aug. 2025
*Forward Deployed Engineer* *Shenzhen, China*

- Fine-tuned Qwen3 models with full-parameter, LoRA, and QLoRA SFT in LLaMA-Factory; served with vLLM.
- Built a knowledge-distillation pipeline with distributed DeepSeek-R1 as teacher to fine-tune Llama-3.2-3B, raising GSM8K accuracy from 70% to 77%.
- Developed a cloud GRPO reinforcement-learning workflow with veRL and Qwen3-8B, packaged as reusable skills for data preparation, distributed training, and serving.

**Elsevier** Nov. 2024 – Apr. 2025
*Data Science Intern* *Beijing, China*

- Analysed large-scale scholar and publication data to identify research trends in computer science, environmental science, and materials science; supported THE World University Subject Rankings.

## Open-Source Contributions

- **Terminal-Bench Science** (task author, PR #2254, proposal approved and domain review passed, final review pending): an operations-research task in which an agent must plan admissions, rooms, and discharges for 450–500 patients across 70–80 beds under 13 constraints, estimate readmission-risk curves from confounded records, and hand in a branch-and-bound certificate within 1% of optimum (or an infeasibility certificate) that the verifier re-solves leaf by leaf. Frontier agents (Claude Opus 5.5, GPT-6) pass at most 2 of 4 runs.

## Selected Projects

- **trustingdata.ai**: platform for human-in-the-loop multimodal and agentic AI evaluation (Stanford GSB AIM-Lab).
- **quantdesk**: LLM-driven alpha-mining loop for WorldQuant BRAIN with Thompson sampling over archived candidates and an auditable lineage archive.

## Honors & Awards

- Meritorious Winner, Mathematical Contest in Modeling (MCM), COMAP, USA, 2023.
- Honorable Mention, Mathematical Contest in Modeling (MCM), COMAP, USA, 2022.

## Technical Skills

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| **Languages** | Python, C++, Go |
| **ML & LLMs** | PyTorch, vLLM, LLaMA-Factory, veRL, LangGraph; SFT, LoRA/QLoRA, GRPO/RLHF, RAG; LLM & agent evaluation, benchmark design |
| **Inference optimization** | Quantization, speculative decoding, constrained decoding |
| **Systems** | Git, Docker, Kubernetes, CI/CD, CUDA, Google Cloud |